HP to kW Chart

Standard motor ratings converted both ways: NEMA horsepower sizes in kilowatts, and IEC kilowatt sizes in horsepower.

One horsepower is 0.7457 kW, so kW is horsepower times 0.746 and horsepower is kW times 1.341. A 50 hp motor is 37.3 kW. This is output power at the shaft; the electrical input is higher, by the motor's losses.

NEMA horsepower ratings in kW

hp	kW	hp	kW	hp	kW
0.5	0.373	15	11.2	125	93.2
0.75	0.559	20	14.9	150	112
1	0.746	25	18.6	200	149
1.5	1.12	30	22.4	250	186
2	1.49	40	29.8	300	224
3	2.24	50	37.3	350	261
5	3.73	60	44.7	400	298
7.5	5.59	75	55.9	450	336
10	7.46	100	74.6	500	373

Shaft output power.

IEC kW ratings in horsepower

kW	hp	kW	hp	kW	hp
0.37	0.496	7.5	10.1	75	101
0.55	0.738	11	14.8	90	121
0.75	1.01	15	20.1	110	148
1.1	1.48	18.5	24.8	132	177
1.5	2.01	22	29.5	160	215
2.2	2.95	30	40.2	200	268
3	4.02	37	49.6	250	335
4	5.36	45	60.3	315	422
5.5	7.38	55	73.8		

Shaft output power. The nearest NEMA size is usually the next one above the figure shown.

How the values are calculated

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kW = hp × 0.7457
hp = kW × 1.341
Input kW = output kW / efficiency
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Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

The conversion is exact for mechanical horsepower. It gives shaft output, not electrical input, and it does not give current: full-load amps depend on voltage, efficiency and power factor, and for sizing conductors and protection the NEC requires the values in its motor tables, not a calculated figure. Metric horsepower, used on some European nameplates, is slightly smaller at 0.7355 kW.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/hp-to-kw-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.